

L18



L18 : The Primitive Root Theorem (1)

Continued...

Lemma 5.14: Let p be an odd prime number. Then there exists a primitive root r modulo p such that

$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$

Proof: Let r be any primitive root modulo p . If $r^{p-1} \not\equiv 1 \pmod{p^2}$, then the proof is complete. Assume that $r^{p-1} \equiv 1 \pmod{p^2}$. Set $r' := r + p$. Since $r' \equiv r \pmod{p}$, we have that r' is a primitive root modulo p . We will prove that $(r')^{p-1} \not\equiv 1 \pmod{p^2}$. Assume by way of contradiction, that $(r')^{p-1} \equiv 1 \pmod{p^2}$. We have

$$\begin{aligned} (r')^{p-1} &= (r+p)^{p-1} \\ &= r^{p-1} + (p-1)r^{p-2}p + \binom{p-1}{2}r^{p-3}p^2 \\ &\quad + \dots + \binom{p-1}{p-2}r^{p-2}p^{p-1} + p^{p-1}. \end{aligned}$$

$$\equiv r^{p-1} + (p-1)r^{p-2}p \pmod{p^2} \quad (2)$$

$$\equiv r^{p-1} + p^2 r^{p-2} - pr^{p-2} \pmod{p^2}$$

$$\equiv r^{p-1} - pr^{p-2} \pmod{p^2}$$

$$\equiv 1 - pr^{p-2} \pmod{p^2} \quad (\text{since } r^{p-1} \equiv 1 \pmod{p^2})$$

Since $(r')^{p-1} \equiv 1 \pmod{p^2}$ we have that

$$1 - pr^{p-2} \equiv 1 \pmod{p^2},$$

and thus

$$pr^{p-2} \equiv 0 \pmod{p^2}.$$

So $p^2 \mid pr^{p-2}$, which implies that $p \mid r^{p-2}$, and thus $p \mid r$. This is a contradiction since r is a primitive root modulo p . Thus $(r')^{p-1} \not\equiv 1 \pmod{p^2}$, and the proof is complete. $\square$

Corollary 5.15: Let p be an odd prime number and r be a primitive root modulo p . Then either r or $r+p$ is a primitive root modulo p^2 .

Proof: Since r is a primitive root modulo p , (3) we have $\text{ord}_p r = \phi(p) = p-1$. Let $n = \text{ord}_{p^2} r$.

Then

$$r^n \equiv 1 \pmod{p^2},$$

and, by Proposition 5.1,

$$n \mid \phi(p^2) = p(p-1). \quad (1)$$

Also, $r^n \equiv 1 \pmod{p^2}$ implies that $r^n \equiv 1 \pmod{p}$, from which

$$\phi(p) = p-1 \mid n, \quad (2)$$

by Proposition 5.1. Now, (1) and (2) imply that either $n = p-1$ or $n = p(p-1) = \phi(p^2)$. If $n = p(p-1) = \phi(p^2)$, then r is a primitive root modulo p^2 , and the proof is complete.

If $n = p-1$, put $r' = r+p$. Since $r' \equiv r \pmod{p}$, r' is a primitive root modulo p , and by a similar reasoning to the above, we have $\text{ord}_{p^2} r' = p-1$ or $\text{ord}_{p^2} r' = p(p-1)$.

The proof of Lemma 5.14 shows that

$\text{ord}_{p^2} r' \neq p-1$. Thus $\text{ord}_{p^2} r' = p(p-1) = \phi(p^2)$, (4)
and the proof is complete. $\square$

Lemma 5.16: Let p be an odd prime and let r be a primitive root modulo p such that $r^{p-1} \not\equiv 1 \pmod{p^2}$. If $m \in \mathbb{Z}$ and $m \geq 2$, then

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m}.$$

Proof: We use induction on m . If $m=2$, then this is the hypothesis of the Theorem. Assume that $k \geq 2$, and that the desired statement holds for $m=k$ so that $r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$. We must show that $r^{p^{k-1}(p-1)} \not\equiv 1 \pmod{p^{k+1}}$, so that the desired statement holds for $m=k+1$. Since $(r, p) = 1$, we have that $(r, p^{k-1}) = 1$, and by Euler's Theorem,

$$r^{p^{k-2}(p-1)} = r^{\phi(p^{k-1})} \equiv 1 \pmod{p^{k-1}}.$$

Thus $p^{k-1} \mid r^{p^{k-2}(p-1)} - 1$, and so there exists $a \in \mathbb{Z}$ such that $r^{p^{k-2}(p-1)} - 1 = ap^{k-1}$.

(5)

Furthermore, $p \nmid a$, for otherwise $r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}$, contrary to the induction hypothesis. We raise each side of the equation

$$r^{p^{k-2}(p-1)} = 1 + ap^{k-1}$$

to the p th power. We obtain

$$\begin{aligned} r^{p^{k-1}(p-1)} &= (1 + ap^{k-1})^p \\ &= 1 + p(ap^{k-1}) + \binom{p}{2}(ap^{k-1})^2 \\ &\quad + \dots + \binom{p}{p-1}(ap^{k-1})^{p-1} + (ap^{k-1})^p \\ &\equiv 1 + ap^k \pmod{p^{k+1}} \\ &\not\equiv 1 \pmod{p^{k+1}}, \end{aligned}$$

Since $p \nmid a$. The proof is complete.

□

Proposition 5.17: Let p be an odd prime number and let m be a positive integer. Then there exists a primitive root modulo p^m .

Proof: The proof for $m=1$ is a consequence of Corollary 5.10. By Lemma 5.14 and Lemma 5.16, there exists a primitive root r modulo p such that

$$r^{p^{m-2}(p-1)} \not\equiv 1 \pmod{p^m} \quad (3)$$

For all $m \in \mathbb{Z}$ with $m \geq 2$. We show that r is a primitive root modulo p^m . Let $n = \text{ord}_{p^m} r$. Then

$$r^n \equiv 1 \pmod{p^m},$$

and, by Proposition 5.1,

$$n \mid \phi(p^m) = p^{m-1}(p-1). \quad (4)$$

Since $r^n \equiv 1 \pmod{p^m}$ implies that $r^n \equiv 1 \pmod{p}$ we have that

$$\phi(p) = p-1 \mid n \quad (5)$$

by Proposition 5.1. Now, (4) and (5) imply that $n = p^k(p-1)$ for some integer k between 0 and $m-1$ inclusive. If $0 \leq k \leq m-2$,

(7)

then

$$\begin{aligned}
 r^{p^{m-2}(p-1)} &= (r^{p^k(p-1)})^{p^{m-2-k}} \\
 &= (r^n)^{p^{m-2-k}} \\
 &\equiv 1^{p^{m-2-k}} \pmod{p^m} \\
 &\equiv 1 \pmod{p^m},
 \end{aligned}$$

which contradicts (3). Thus $k = m-1$, and

$$\text{ord}_{p^m} r = n = p^{m-1}(p-1) = \phi(p^m),$$

from which r is a primitive root modulo p^m , as desired. $\square$

Corollary 5.18: Let p be an odd prime number and let m be a positive integer. Then there exists a primitive root modulo $2p^m$.

Proof: Let r be a primitive root modulo p^m . Without loss of generality, we may assume that r is odd (if r is even, then $r+p^m$ is odd and a primitive root modulo p^m). So $(r, 2p^m) = 1$. Let $n = \text{ord}_{(2p^m)} r$. Then

$r^n \equiv 1 \pmod{2p^m}$, and, by Proposition 5.1, (8)

$$n \mid \phi(2p^m) = \phi(2) \phi(p^m) = \phi(p^m). \quad (6)$$

Also $r^n \equiv 1 \pmod{2p^m}$ implies that

$$r^n \equiv 1 \pmod{p^m} \quad (7), \text{ from which}$$

$$\phi(p^m) \mid n$$

by Proposition 5.1. Together, (6) and (7) imply that

$$\text{ord}_{(2p^m)} r = n = \phi(p^m) = \phi(2p^m),$$

from which r is a primitive root as desired. $\square$

Theorem 5.19 (Primitive Root Theorem):

Let n be a positive integer. Then a primitive root modulo n exists if and only if n is equal to $1, 2, 4, p^m$ or $2p^m$, where p is an odd prime number and m is a positive integer.

Proof: ($\Rightarrow$ direction) This is the last statement

of Corollary 5.13.

⑨

($\Leftarrow$ direction) It is easily checked that 1 is a primitive root modulo 1, 1 is a primitive root modulo 2, and 3 is a primitive root modulo 4. Proposition 5.17 and Corollary 5.18 prove the existence of primitive roots modulo p^m and $2p^m$.

□

Exercise : • For each positive integer $m \geq 1$, find a primitive root for the modulus 11^m .
• Do the same for the modulus 17^m .